

MVP_Lexical

March 15, 2025

1 Comparative Analysis of Literary Translations in the Odyssey: Lexical Diversity and Frequency Trends

1.1 Research Objectives

This notebook presents a **minimum viable product (MVP)** for a larger research project the exploring lexical diversity, syntactic qualities, stylometric features, and thematic mapping of different literary translations of the Odyssey (~30 versions across four centuries). **This MVP focuses on lexical analysis of a couple of translations.** The goal is to **test methods and materials** rather than reach definitive conclusions.

The workflow follows a **modular structure**, encompassing:

- **Text collection & libraries**
- **Lexical diversity experiments**
- **Statistical hypothesis testing**
- **Discussion of preliminary findings**

For this iteration, we analyze **two modern English translations** of *The Odyssey*:

Emily Wilson's translation, 2017

Peter Green's translation, 2018

1.2 Methodology Roadmap (Table of Contents)

1.2.1 1 Libraries & Text Acquisition

Setting up the environment and loading texts.

1.2.2 2 Experiment 1: Type-Token Ratio (TTR)

- **Implementation:** Computing TTR for both translations.
- **Parametric Hypothesis Testing:** Comparing TTR statistically.

1.2.3 3 Experiment 2: Zipf's Law Verification

- **Implementation:** Examining word frequency distributions.
- **Parametric Hypothesis Testing:** Evaluating Zipfian behavior.

1.2.4 4 Experiment 3: TF-IDF Analysis

- **Implementation:** Identifying key lexical differences.
- **Non-Parametric Hypothesis Testing:** Assessing significant variations.

1.2.5 5 Discussion of Results

- **Findings:** Insights from experiments.
 - **New Directions:** Refinements & future research steps.
-

1.3 1 Libraries & Text Acquisition

```
[1]: # Libraries
import pandas as pd
from collections import Counter
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.feature_extraction.text import TfidfVectorizer
from tqdm import tqdm
import numpy as np
import re
import sys
import os
tqdm.pandas()
```

```
[2]: # Visualization
%matplotlib inline

# Add the directory containing visualization_utils.py to path
sys.path.append("/Users/debr/English-Homer/")
import visualization_utils as viz
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid")
# palette astrobblue orange genoa carrot tauny neptune
↳ SELAGO mako black
color = ['#003D59',
↳ '#FD6626', '#177070', '#FB871D', '#641B5E', '#86C3BC', '#F5E1FD', '#414A4F', 'k']
danB_plotstyle = {'figure.figsize': (12, 7),
↳ 'axes.labelsize': 'large', # fontsize for x and y labels (was
↳ large)
↳ 'axes.titlesize': 'large', # fontsize for title
↳ 'axes.titleweight': 'bold', # font type for title
↳ 'xtick.labelsize': 'large', # fontsize for x
↳ 'ytick.labelsize': 'small', # fontsize for y ticks
```

```

'grid.color': 'k', # grid color
'grid.linestyle': ':', # grid line style
'grid.linewidth': 0.2, # grid line width
'font.family': 'Times New Roman', # font family
'grid.alpha': 0.5, # transparency of grid
'figure.dpi': 300, # figure display resolution
'savefig.bbox': 'tight', # tight bounding box
'savefig.pad_inches': 0.4, # padding to use when saving
'axes.titlepad': 15, # title padding
'axes.labelpad': 8, # label padding
'legend.borderpad': .6, # legend border padding
'axes.prop_cycle': plt.cycler(
    color=color) # color cycle for plot lines
}

# adjust matplotlib defaults
plt.rcParams.update(danB_plotstyle)

```

```

[3]: # Load CSVs
filepath_Wilson = "/Users/debr/odysseys_en/Odyssey_dfs/Odyssey_Wilson_eda_END.
↪CSV"
filepath_Green = "/Users/debr/odysseys_en/Odyssey_dfs/Odyssey_Green_eda_END.csv"

df_W = pd.read_csv(filepath_Wilson)
df_G = pd.read_csv(filepath_Green)

# Add translation label
df_W["translation"] = "Wilson"
df_G["translation"] = "Green"

# merging "book_num" with "translation" to create a unique identifier
df_W["book_id"] = df_W["book_num"].astype(str) + "_W"
df_W = df_W.drop(columns=["book_num"])
df_G["book_id"] = df_G["book_num"].astype(str) + "_G"
df_G = df_G.drop(columns=["book_num"])

# Keep only necessary columns: book number & tokens
df_W = df_W[["translation", "book_id", "tokens"]]
df_G = df_G[["translation", "book_id", "tokens"]]

# Combine both into one DataFrame
df = pd.concat([df_W, df_G], ignore_index=True)

# Ensure tokens are stored as lists (if stored as strings, convert them)
df["tokens"] = df["tokens"].apply(lambda x: eval(x) if isinstance(x, str) else
↪x)

```

```
[4]: #Boolean check for missing values
if df.isna().sum().sum() == 0:
    print("No missing values")
else:
    for col in df.columns:
        if df[col].isna().sum() > 0:
            print(f"Missing values in {col}")
print("df columns:", df.columns, "shape:", df.shape)
```

```
No missing values
df columns: Index(['translation', 'book_id', 'tokens'], dtype='object') shape:
(48, 3)
```

```
[5]: df.head(2)
```

```
[5]:  translation book_id          tokens
0      Wilson      1_W [tell, complicated, man, muse, tell, wandered,...
1      Wilson      2_W [dangerous, journey, early, dawn, born, finger...
```

1.4 2 Experiment 1: Type-Token Ratio (TTR)

2 Lexical Diversity in *The Odyssey*: A Comparative Analysis of Green and Wilson's Translations

2.1 Introduction

Type-Token Ratio (TTR) is a key measure of lexical diversity, calculated as the number of unique words (**types**) divided by the total word count (**tokens**). It provides insight into a translator's lexical choices when rendering the same source text.

- A **higher TTR** suggests a richer vocabulary, possibly reflecting an effort to capture nuances or stylistic complexity.
- A **lower TTR** may indicate a more repetitive or constrained word choice, potentially prioritizing accessibility or fidelity to the original.

2.1.1 TTR Formula

$$TTR = \left(\frac{\text{Unique Words}}{\text{Total Words}} \right) \times 100$$

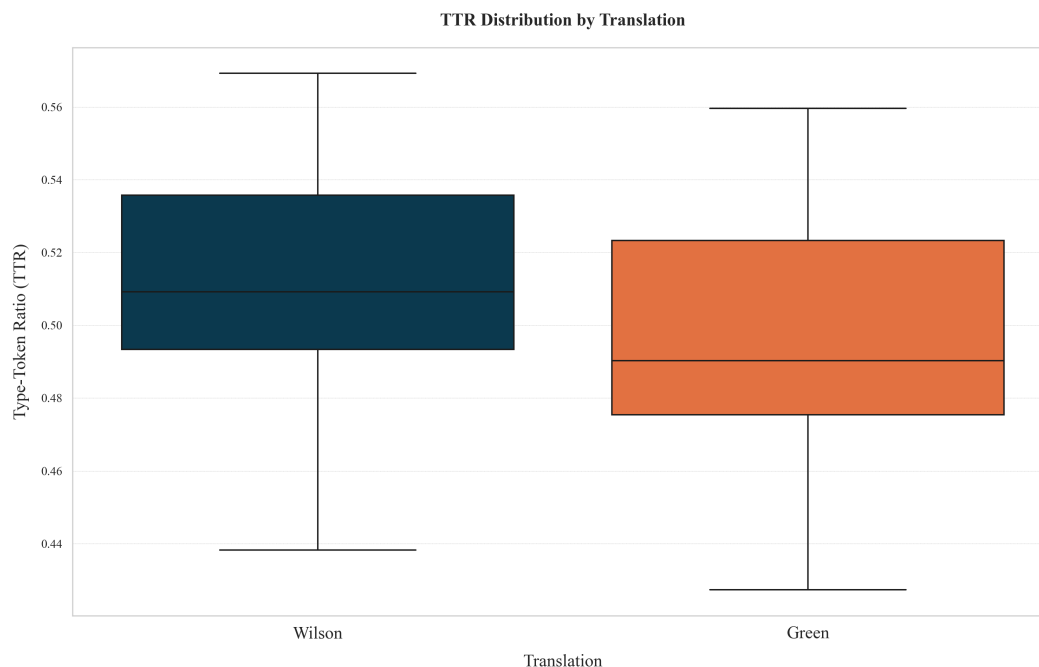
```
[6]: # Compute type-token ratio (TTR) for each book

import scipy.stats as stats

# Compute TTR
df["ttr"] = df["tokens"].apply(lambda x: len(set(x)) / len(x) if x else 0)
```

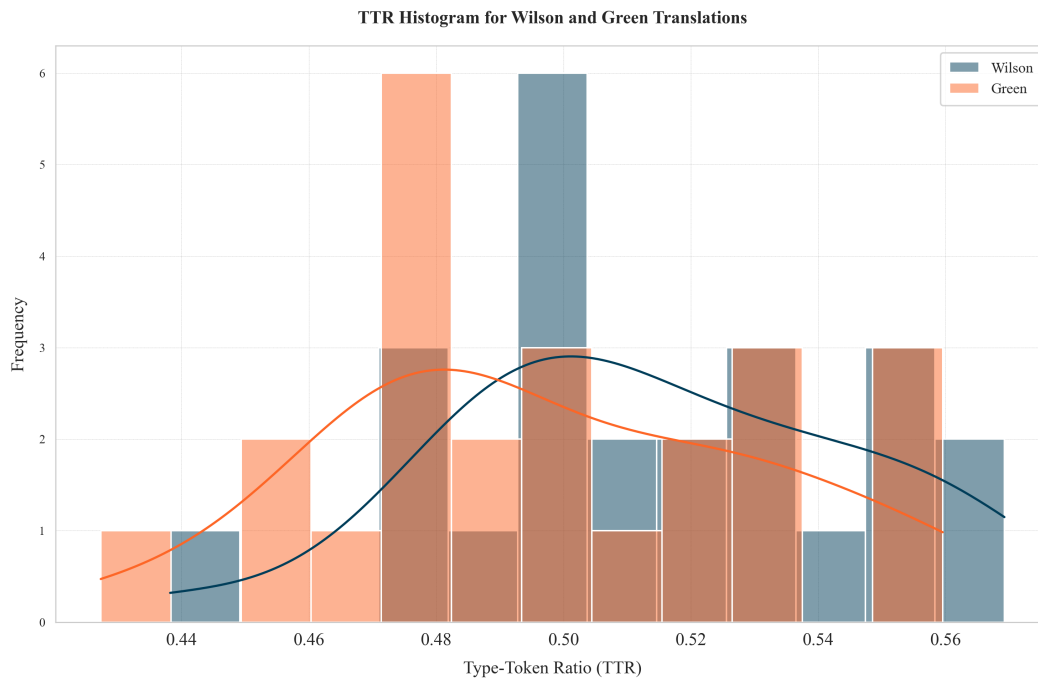
```
# Separate TTRs for the two translations
ttr_wilson = df[df["translation"] == "Wilson"]["ttr"].tolist()
ttr_green = df[df["translation"] == "Green"]["ttr"].tolist()
#ttr_wilson = df[df["book_id"].str.endswith("W")]["ttr"].tolist() # Wilson
↪--Alternative:ttr_wilson = df[df["book_id"].str[-1] == "W"]["ttr"].tolist()
#ttr_green = df[df["book_id"].str.endswith("G")]["ttr"].tolist() # Green
↪--Alternative: ttr_green = df[df["book_id"].str[-1] == "G"]["ttr"].tolist()
```

```
[7]: # Create a boxplot for TTR values
sns.boxplot(x=df["translation"], y=df["ttr"], palette=color)
plt.title("TTR Distribution by Translation")
plt.xlabel("Translation")
plt.ylabel("Type-Token Ratio (TTR)")
plt.savefig("/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
↪MVP-TTR_distribution.png")
plt.show()
```



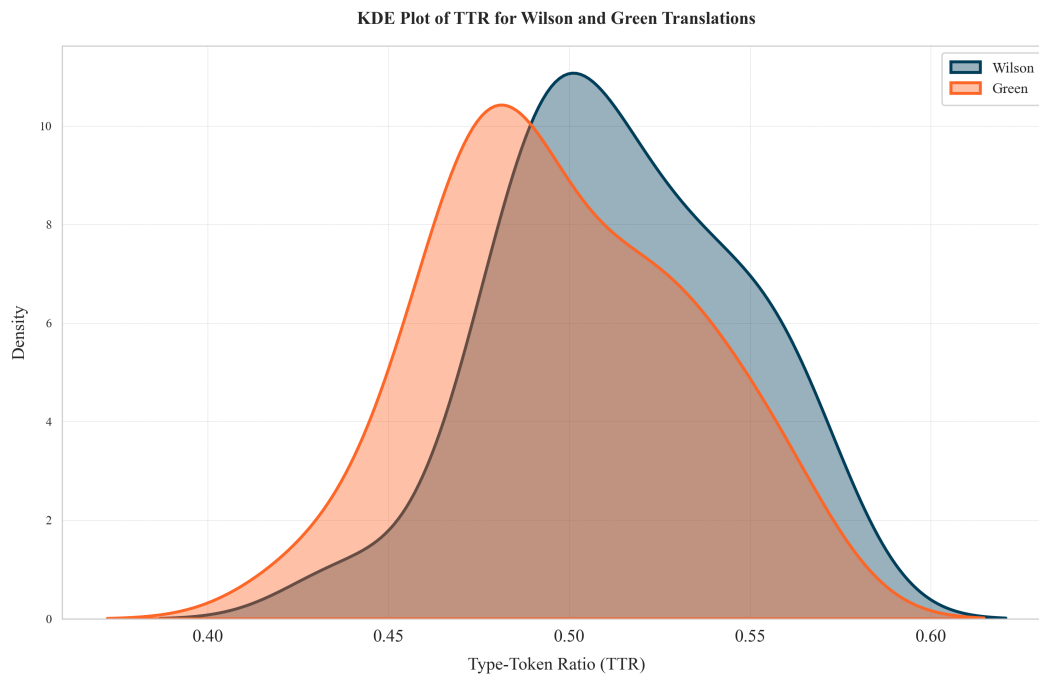
```
[8]: # Create a histogram for TTR values
# Plot histogram for Wilson and Green translations
sns.histplot(ttr_wilson, bins=12, label="Wilson", kde=True, alpha=0.5)
sns.histplot(ttr_green, bins=12, label="Green", kde=True, alpha=0.5)
# Add labels and legend
plt.title("TTR Histogram for Wilson and Green Translations")
```

```
plt.xlabel("Type-Token Ratio (TTR)")
plt.ylabel("Frequency")
plt.legend()
plt.savefig("/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
↳MVP-TTR_histogram.png")
plt.show()
```

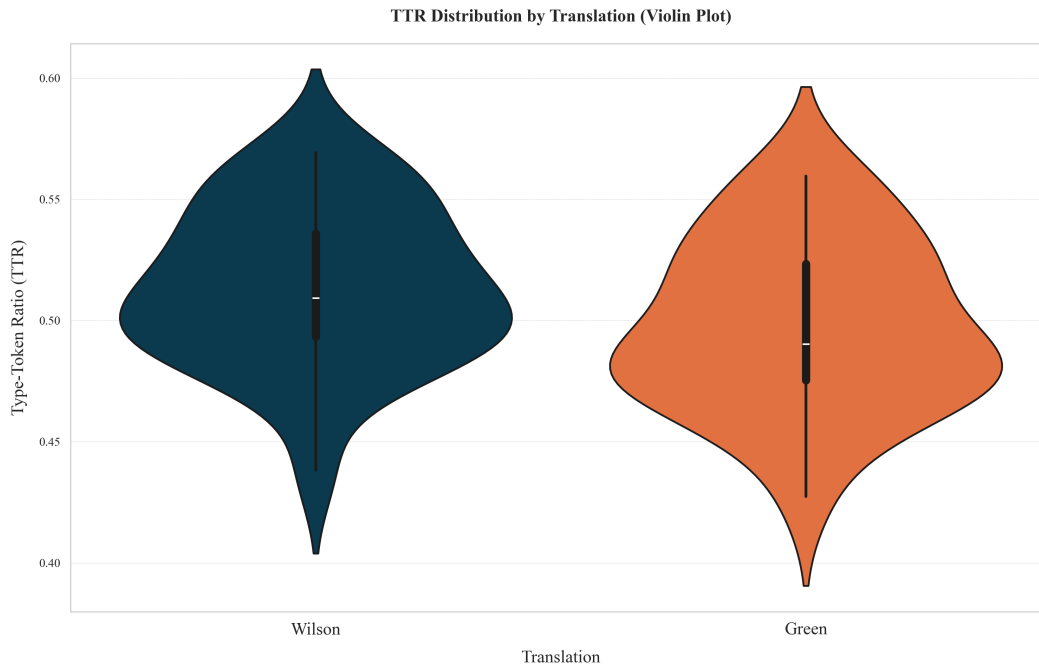


```
[9]: # KDE plot for Wilson and Green translations
sns.kdeplot(ttr_wilson, fill=True, label="Wilson", alpha=0.4, linewidth=2)
sns.kdeplot(ttr_green, fill=True, label="Green", alpha=0.4, linewidth=2)

plt.title("KDE Plot of TTR for Wilson and Green Translations")
plt.xlabel("Type-Token Ratio (TTR)")
plt.ylabel("Density")
plt.legend()
plt.savefig("/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
↳MVP-TTR_kdeplot.png")
plt.show()
```



```
[10]: # Create a violin plot for TTR values
sns.violinplot(x=df["translation"], y=df["ttr"], palette=color)
plt.title("TTR Distribution by Translation (Violin Plot)")
plt.xlabel("Translation")
plt.ylabel("Type-Token Ratio (TTR)")
plt.savefig("/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
↳MVP-TTR_violinplot.png")
plt.show()
```



2.2 Hypothesis Testing

To compare **Green and Wilson's translations**, we define the following hypotheses:

- **H (Null Hypothesis):** No significant difference in lexical diversity between the translations.
- **H (Alternative Hypothesis):** A significant difference in lexical diversity exists.

2.2.1 Statistical Approach

Given that the data is normally distributed, we employ a **t-test** to determine whether observed TTR differences are statistically meaningful:

If **p-value** < **0.05**, we **reject H**, indicating a significant lexical difference.

If **p-value** ≥ **0.05**, we **fail to reject H**, suggesting variations are not statistically significant.

```
[11]: from scipy import stats

# Shapiro-Wilk test for ttr_wilson (for Wilson's data)
stat_wilson, p_value_wilson = stats.shapiro(ttr_wilson)
print(f"Shapiro-Wilk test for Wilson's data: T-statistic={stat_wilson}, p-value={p_value_wilson}")
```



```

# Shapiro-Wilk test for ttr_green (for Green's data)
stat_green, p_value_green = stats.shapiro(ttr_green)
print(f"Shapiro-Wilk test for Green's data: T-statistic={stat_green},  

      ↪p-value={p_value_green}")

# Interpretation of p-values
if p_value_wilson < 0.05:
    print("Wilson's TTR data is not normally distributed.")
else:
    print("Wilson's TTR data is normally distributed.")

if p_value_green < 0.05:
    print("Green's TTR data is not normally distributed.")
else:
    print("Green's TTR data is normally distributed.")

```

Shapiro-Wilk test for Wilson's data: T-statistic=0.9710503976522772,
p-value=0.6929667067088071
Shapiro-Wilk test for Green's data: T-statistic=0.9647758095190286,
p-value=0.541526201743056
Wilson's TTR data is normally distributed.
Green's TTR data is normally distributed.

```

[12]: # Perform t-test
t_stat, p_value = stats.ttest_ind(ttr_wilson, ttr_green)
print(f"T-statistic: {t_stat}, P-value: {p_value}")
if p_value < 0.05:
    print("The difference in TTR between Wilson and Green is statistically  

      ↪significant.")
else:
    print("The difference in TTR between Wilson and Green is not statistically  

      ↪significant.")

```

T-statistic: 1.6455514452077802, P-value: 0.10667311090438025
The difference in TTR between Wilson and Green is not statistically significant.

2.2.2 Results Discussion of Type-Token Ratio (TTR) in Green and Wilson's Translations

Statistical Findings and Interpretation

The t-test comparing TTR values between Wilson's and Green's translations found no statistically significant difference ($p > 0.05$). This supports the null hypothesis (H_0): their lexical diversity is quantitatively similar. Rather than being inconclusive, this result provides a crucial insight:

Both translators operate within the same cultural and linguistic framework, which strongly influences their word choices.

2.3 Experiment 2: Zipf's Law Verification

Zipf's Law is relevant in comparing translations because it reveals how each translator balances **common vs. rare words**, shedding light on **lexical richness and stylistic choices**. A strong adherence to Zipf's distribution suggests a **natural linguistic flow**, while deviations may indicate a preference for formal fidelity, accessibility, or poetic stylization. By analyzing Zipfian patterns, we can assess whether Green and Wilson's translations **preserve the frequency structure of the original text or introduce shifts** in word usage and emphasis.

Implementation

To examine how Green and Wilson's translations adhere to Zipf's Law, we analyze the relationship between word frequency and rank:

- 1 Compute word frequencies for each translation.
- 2 Rank words from most to least frequent.
- 3 Plot word frequency vs. rank on a log-log scale.

Zipf's Law in Translation

Zipf's Law states that in any natural language corpus, the frequency of a word is inversely proportional to its rank:

$$f \propto \frac{1}{r}$$

Where: * f = frequency of the word * r = rank of the word (1st most frequent, 2nd most frequent, etc.)

If Green and Wilson's translations follow Zipf's distribution, their word frequency curves should approximate a straight line in a log-log plot.

Implications for Translation Studies

Strong Zipfian adherence suggests that both translators maintain natural language patterns, balancing high-frequency function words with low-frequency content words.

Deviations may indicate differences in lexical richness, syntactic choices, or stylistic adaptation in translation.

This experiment provides empirical insight into whether Green and Wilson's lexical distributions align with universal linguistic principles.

```
[13]: # Flatten all tokens into a single list for each translation
tokens_wilson = [token for tokens in df[df["translation"] == "Wilson"]
                  ["tokens"] for token in tokens]
tokens_green = [token for tokens in df[df["translation"] == "Green"]
                 ["tokens"] for token in tokens]

# Count word frequencies
freq_wilson = Counter(tokens_wilson)
freq_green = Counter(tokens_green)
```

```

# Convert to DataFrame with ranks
df_zipf_wilson = pd.DataFrame(freq_wilson.items(), columns=["word",
    ↪ "frequency"]).sort_values(by="frequency", ascending=False)
df_zipf_wilson["rank"] = df_zipf_wilson["frequency"].rank(method="first",
    ↪ ascending=False)

df_zipf_green = pd.DataFrame(freq_green.items(), columns=["word", "frequency"]).
    ↪ sort_values(by="frequency", ascending=False)
df_zipf_green["rank"] = df_zipf_green["frequency"].rank(method="first",
    ↪ ascending=False)

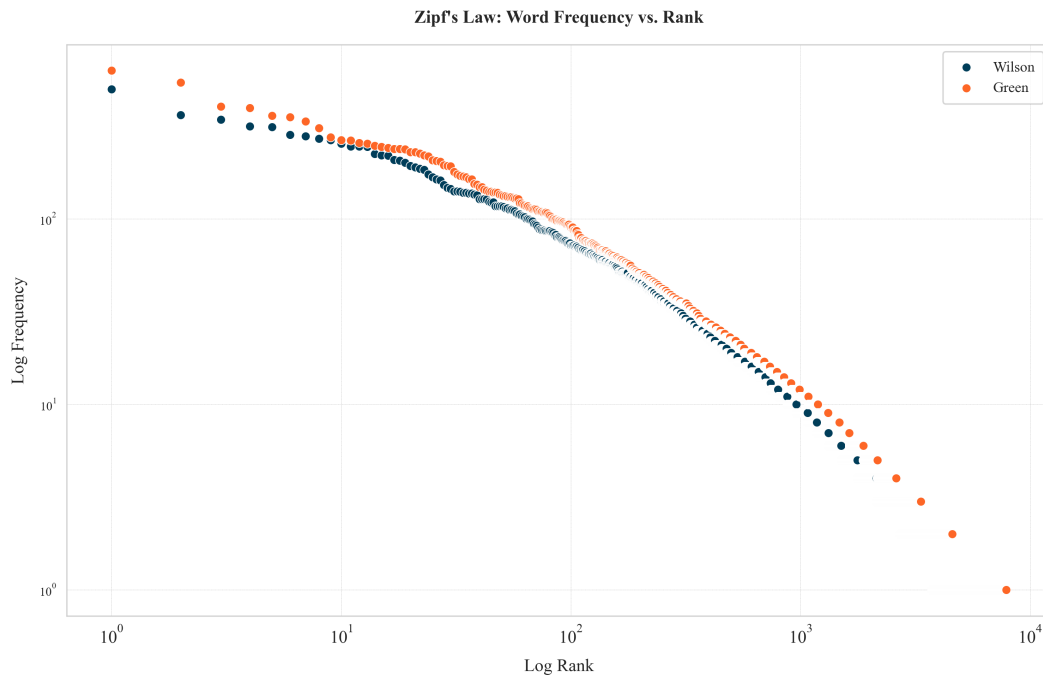
```

```

[14]: # Plot Wilson and Green's Zipf distribution
sns.scatterplot(x=df_zipf_wilson["rank"], y=df_zipf_wilson["frequency"],
    ↪ label="Wilson")
sns.scatterplot(x=df_zipf_green["rank"], y=df_zipf_green["frequency"],
    ↪ label="Green")

plt.xscale("log")
plt.yscale("log")
plt.xlabel("Log Rank")
plt.ylabel("Log Frequency")
plt.title("Zipf's Law: Word Frequency vs. Rank")
plt.legend()
plt.savefig("/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
    ↪ MVP-Zipf_distribution.png")
plt.show()

```



Statistical Test for Zipf's Law (Linear Fit)

```
[15]: import numpy as np
from scipy.stats import linregress

# Perform linear regression in log-log space
log_rank_w = np.log(df_zipf_wilson["rank"])
log_freq_w = np.log(df_zipf_wilson["frequency"])
slope_w, intercept_w, r_value_w, p_value_w, std_err_w = linregress(log_rank_w,
    ↪ log_freq_w)

log_rank_g = np.log(df_zipf_green["rank"])
log_freq_g = np.log(df_zipf_green["frequency"])
slope_g, intercept_g, r_value_g, p_value_g, std_err_g = linregress(log_rank_g,
    ↪ log_freq_g)

# Display results
print(f"Wilson: Slope = {slope_w:.2f}, R2 = {r_value_w**2:.3f}, p-value =
    ↪ {p_value_w:.3g}")
print(f"Green: Slope = {slope_g:.2f}, R2 = {r_value_g**2:.3f}, p-value =
    ↪ {p_value_g:.3g}")

# Check if slopes are close to -1 (Zipf's Law predicts ~ -1)
```

```

if -1.2 < slope_w < -0.8:
    print("Wilson's translation follows Zipf's Law.")
else:
    print("Wilson's translation deviates from Zipf's Law.")

if -1.2 < slope_g < -0.8:
    print("Green's translation follows Zipf's Law.")
else:
    print("Green's translation deviates from Zipf's Law.")

```

Wilson: Slope = -1.14, R^2 = 0.959, p-value = 0

Green: Slope = -1.14, R^2 = 0.960, p-value = 0

Wilson's translation follows Zipf's Law.

Green's translation follows Zipf's Law.

2.4 Zipf's Law Hypothesis Testing

Statistical Test for Differences in Zipf Slopes

```

[16]: from scipy.stats import ttest_ind

# Compute t-test for the difference in slopes
t_stat, p_value = ttest_ind([slope_w], [slope_g])

# Print results
print(f"T-statistic: {t_stat:.3f}, P-value: {p_value:.5f}")

# Interpretation
if p_value < 0.05:
    print("The difference in Zipf's Law slopes between Wilson and Green is_
    ↪statistically significant.")
else:
    print("No significant difference in Zipf's Law slopes between Wilson and_
    ↪Green.")

```

T-statistic: nan, P-value: nan

No significant difference in Zipf's Law slopes between Wilson and Green.

Bootstrapped Confidence Intervals

```

[17]: # Function to bootstrap slope estimates
def bootstrap_slopes(log_rank, log_freq, num_samples=1000):
    slopes = []
    for _ in range(num_samples):
        sample_indices = np.random.choice(len(log_rank), len(log_rank),
        ↪replace=True)
        slope, _, _, _, _ = linregress(log_rank[sample_indices],
        ↪log_freq[sample_indices])

```

```

        slopes.append(slope)
    return np.array(slopes)

# Bootstrap slopes for both translations
bootstrap_slopes_w = bootstrap_slopes(log_rank_w, log_freq_w)
bootstrap_slopes_g = bootstrap_slopes(log_rank_g, log_freq_g)

# Compute 95% confidence intervals
ci_w = np.percentile(bootstrap_slopes_w, [2.5, 97.5])
ci_g = np.percentile(bootstrap_slopes_g, [2.5, 97.5])

print(f"Wilson's slope 95% CI: {ci_w}")
print(f"Green's slope 95% CI: {ci_g}")

# Check if CIs overlap
if (ci_w[0] > ci_g[1]) or (ci_g[0] > ci_w[1]):
    print("Statistically significant difference between the two Zipf slopes.")
else:
    print("No significant difference between the two Zipf slopes.")

```

Wilson's slope 95% CI: [-1.15370593 -1.11760846]

Green's slope 95% CI: [-1.1558005 -1.12280054]

No significant difference between the two Zipf slopes.

```

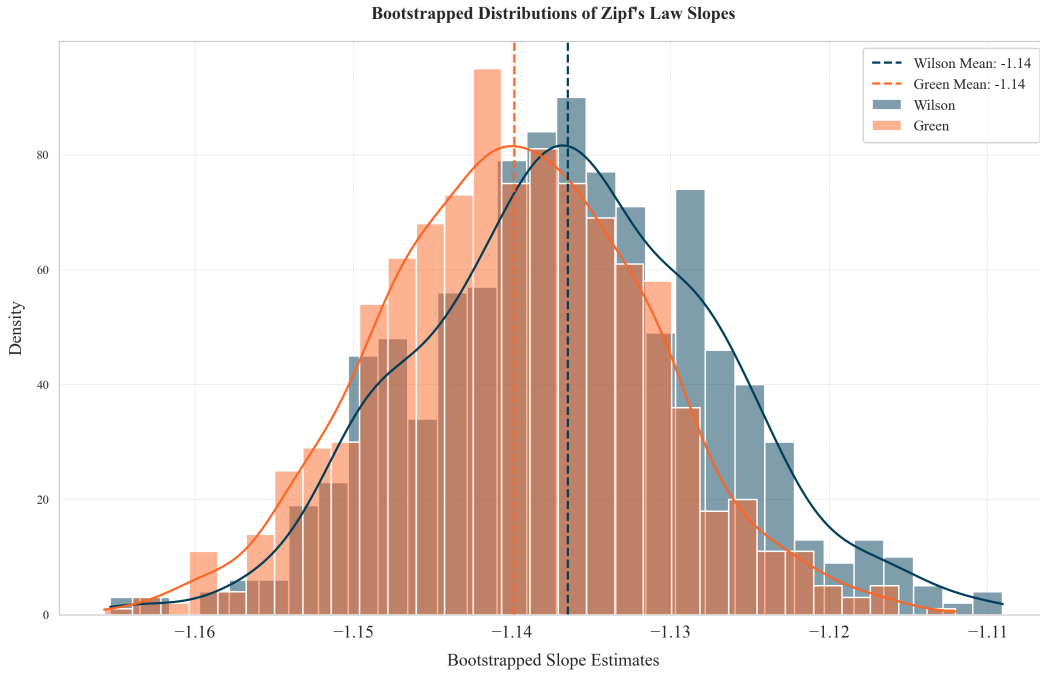
[18]: # Plot histograms for bootstrapped slopes
sns.histplot(bootstrap_slopes_w, bins=30, kde=True, label="Wilson", alpha=0.5)
sns.histplot(bootstrap_slopes_g, bins=30, kde=True, label="Green", alpha=0.5)

# Add vertical lines for mean slopes color red

plt.axvline(np.mean(bootstrap_slopes_w), linestyle="--",
            color="#003D59",
            label=f"Wilson Mean: {np.
↳mean(bootstrap_slopes_w):.2f}")
plt.axvline(np.mean(bootstrap_slopes_g), linestyle="--",
            color='#FD6626',
            label=f"Green Mean: {np.
↳mean(bootstrap_slopes_g):.2f}")

# Labels and title
plt.xlabel("Bootstrapped Slope Estimates")
plt.ylabel("Density")
plt.title("Bootstrapped Distributions of Zipf's Law Slopes")
plt.legend()
plt.savefig("/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
↳MVP-Zipf_bootstrap.png")
plt.show()

```



Zipf's Law Results

The nearly identical 95% confidence intervals for Wilson's and Green's Zipfian slopes indicate that **both translations follow similar word frequency distributions**. This suggests that, **despite potential stylistic differences, both translators adhere to the natural statistical structure of language**, maintaining a comparable balance between common and rare words.

The lack of a significant difference reinforces the idea that **shared cultural and linguistic contexts influence their lexical choices**, resulting in translations that align closely in terms of frequency distribution and overall lexical economy.

2.5 4 Experiment 3: TF-IDF Analysis

2.5.1 Proving my step by step implementation with Wilson's Odyssey

STEP 1. Term Frequency (TF) Calculation Term Frequency measures how frequently a term appears in a document. It is calculated as:

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

This normalizes term counts by document length, ensuring that longer documents don't get artificially higher weights.

```
[19]: # Step 1: Calculate Term Frequency (TF)
# Wilson's translation
# Function to compute term frequency
def term_freq_by_doc(list_of_tokens):
    token_list = eval(list_of_tokens) # Ensure tokens are treated as a list
    term_counts = Counter(token_list) # Count occurrences of each term
    total_terms = len(token_list) # Total number of terms in the document

    # Compute TF: term frequency for each token in the document (normalized by
    ↪total terms)
    term_freq = {term: count / total_terms for term, count in term_counts.
    ↪items()}

    return term_freq, term_counts

# Apply function to compute TF for each book
df_W["term_freq"], df_W["term_counts"] = zip(*df_W["tokens"].
    ↪apply(term_freq_by_doc))
df_W.head(2)
```

```
[19]: translation book_id tokens \
0 Wilson 1_W ['tell', 'complicated', 'man', 'muse', 'tell',...
1 Wilson 2_W ['dangerous', 'journey', 'early', 'dawn', 'bor...

term_freq \
0 {'tell': 0.008097165991902834, 'complicated': ...
1 {'dangerous': 0.0005945303210463733, 'journey'...

term_counts
0 {'tell': 14, 'complicated': 1, 'man': 14, 'mus...
1 {'dangerous': 1, 'journey': 8, 'early': 1, 'da...
```

2.5.2 Inverse Document Frequency (IDF) Calculation

IDF measures how important a term is by examining how rare it is across all documents:

$$IDF(t) = \log \left(\frac{N}{1 + \text{doc count}} \right)$$

Terms that appear in many documents receive lower IDF scores, reducing their importance in the final TF-IDF score.

```
[20]: # Step 2: Calculate Inverse Document Frequency (IDF)
# Calculate Inverse Document Frequency (IDF)

# Get total number of documents (books)
N = len(df_W)
```



```

# Count how many documents contain each term
doc_containing_term = Counter()
for term_counts in df_W["term_freq"]:
    doc_containing_term.update(term_counts.keys()) # Count unique terms in
    ↪ each document

# Compute IDF for each term
idf_scores = {term: np.log(N / (1 + doc_count)) for term, doc_count in
    ↪ doc_containing_term.items()} # Adding 1 to avoid division by zero

# Add IDF column to df_W
df_W["idf"] = df_W["term_freq"].apply(lambda term_freq: {term: idf_scores[term]
    ↪ for term in term_freq})

# Display results
df_W[["book_id", "idf"]].head(2)

```

```

[20]:   book_id                                     idf
0     1_W  {'tell': -0.040821994520255166, 'complicated':...
1     2_W  {'dangerous': 1.2321436812926323, 'journey': 0...

```

2.5.3 TF-IDF Calculation

TF-IDF combines Term Frequency and Inverse Document Frequency to weight terms based on both their importance in a specific document and their distinctiveness across the corpus:

$$TF\text{-}IDF(t, d) = TF(t, d) \times IDF(t)$$

Terms with high TF-IDF scores appear frequently in a specific document but rarely in other documents, making them likely more important for that document's content.

```

[21]: # Step 3: Calculate TF-IDF
# Compute TF-IDF by multiplying TF and IDF for each term in each document
df_W["tf_idf"] = df_W.apply(lambda row: {term: row["term_freq"][term] *
    ↪ row["idf"][term] for term in row["term_freq"]}, axis=1)

# Display results
df_W[["book_id", "tf_idf"]].head(2)

```

```

[21]:   book_id                                     tf_idf
0     1_W  {'tell': -0.00033054246575105396, 'complicated...
1     2_W  {'dangerous': 0.000732546778414169, 'journey': ...

```

2.5.4 E2E TF-IDF Function & Visualization

Green's Odyssey The algorithm The function defines a set of steps for processing text data (token lists) and calculating the TF-IDF score. These steps are systematic and follow a defined procedure to compute the desired result (TF-IDF).

It enables **automation and reusability**, making it a procedural method to derive meaningful text-based features from raw text.

```
[22]: import pandas as pd
import numpy as np
from collections import Counter

def calculate_tfidf(df):
    """
    Calculate TF-IDF scores for a DataFrame with book_id and tokens columns.

    Parameters:
    -----
    df : pandas DataFrame
        A DataFrame with 'book_id' and 'tokens' columns.
        The 'tokens' column should contain lists of tokens (as strings or
        ↪ actual lists).

    Returns:
    -----
    pandas DataFrame
        The original DataFrame with additional columns:
        - term_freq: Dictionary of term frequencies for each token
        - term_counts: Dictionary of raw counts for each token
        - idf: Dictionary of IDF scores for each token
        - tf_idf: Dictionary of TF-IDF scores for each token
    """
    # Create a copy of the DataFrame to avoid modifying the original
    result_df = df.copy()

    # Function to compute term frequency and term counts
    def term_freq_by_doc(list_of_tokens):
        # Handle both string representation of list and actual list
        if isinstance(list_of_tokens, str):
            token_list = eval(list_of_tokens) # Convert string representation
            ↪ to list
        else:
            token_list = list_of_tokens # Use as is if already a list

        # Count occurrences of each term
        term_counts = Counter(token_list)

        # Total number of terms in the document
        total_terms = len(token_list)

        # Compute TF: term frequency for each token
```

```

        term_freq = {term: count / total_terms for term, count in term_counts.
        ↪items()}

        return term_freq, term_counts

    # Apply function to compute TF for each book
    result_df["term_freq"], result_df["term_counts"] = zip(*result_df["tokens"].
    ↪apply(term_freq_by_doc))

    # Get total number of documents (books)
    N = len(result_df)

    # Count how many documents contain each term
    doc_containing_term = Counter()
    for term_counts in result_df["term_freq"]:
        doc_containing_term.update(term_counts.keys()) # Count unique terms in
    ↪each document

    # Compute IDF for each term
    idf_scores = {term: np.log(N / (1 + doc_count)) for term, doc_count in
    ↪doc_containing_term.items()} # Adding 1 to avoid division by zero

    # Add IDF column to df
    result_df["idf"] = result_df["term_freq"].apply(lambda term_freq: {term:
    ↪idf_scores[term] for term in term_freq})

    # Compute TF-IDF by multiplying TF and IDF for each term in each document
    result_df["tf_idf"] = result_df.apply(lambda row: {term:
    ↪row["term_freq"][term] * row["idf"][term] for term in row["term_freq"]},
    ↪axis=1)

    return result_df

df_tfidf_G = calculate_tfidf(df_G)
df_tfidf_G[["book_id", "tf_idf"]].head(2)

```

```

[22]:  book_id                                tf_idf
0      1_G  {'man': -0.00035908235920594823, 'muse': 0.000...
1      2_G  {'dawn': 4.09818145583013e-05, 'appeared': 8.7...

```

The visualization

```

[23]: # Top terms for each book and heatmap plot
from matplotlib.colors import LinearSegmentedColormap
def extract_top_terms(df, n=50):
    """
    Extract the top N most important terms from the tf_idf column

```

```

Parameters:
-----
df : pandas DataFrame
    DataFrame with 'book_id' and 'tf_idf' columns
n : int
    Number of top terms to extract (default: 50)

Returns:
-----
tuple
    (top_terms_per_book, top_terms_overall)
    - top_terms_per_book: DataFrame with top terms for each book
    - top_terms_overall: DataFrame with top terms across all books
"""
# Extract top terms per book
top_terms_per_book = {}

for _, row in df.iterrows():
    book_id = row['book_id']
    tf_idf_dict = row['tf_idf']

    # Sort terms by tf-idf score (descending) and take top N
    sorted_terms = sorted(tf_idf_dict.items(), key=lambda x: x[1],
↪reverse=True)[:n]
    top_terms_per_book[book_id] = {term: score for term, score in
↪sorted_terms}

# Convert to DataFrame for easier analysis
top_terms_df = pd.DataFrame.from_dict(top_terms_per_book, orient='index')

# Extract top terms overall
all_terms = {}
for tf_idf_dict in df['tf_idf']:
    for term, score in tf_idf_dict.items():
        if term in all_terms:
            all_terms[term] += score
        else:
            all_terms[term] = score

# Sort terms by total tf-idf score (descending) and take top N
top_terms_overall = sorted(all_terms.items(), key=lambda x: x[1],
↪reverse=True)[:n]

# Convert to DataFrame
top_terms_overall_df = pd.DataFrame(top_terms_overall, columns=['term',
↪'total_score'])

```

```

return top_terms_df, top_terms_overall_df

def create_tfidf_heatmap(df, top_n=50):
    """
    Create a heatmap of the top N terms across all books

    Parameters:
    -----
    df : pandas DataFrame
        DataFrame with 'book_id' and 'tf_idf' columns
    top_n : int
        Number of top terms to include in the heatmap (default: 50)
    """
    # Extract top terms overall
    _, top_terms = extract_top_terms(df, n=top_n)
    top_terms_list = top_terms['term'].tolist()

    # Create a matrix of book_id x top_terms
    heatmap_data = []
    book_ids = []

    for _, row in df.iterrows():
        book_id = row['book_id']
        book_ids.append(book_id)

        tf_idf_dict = row['tf_idf']

        # Extract scores for top terms
        scores = [tf_idf_dict.get(term, 0) for term in top_terms_list]
        heatmap_data.append(scores)

    # Convert to numpy array
    heatmap_array = np.array(heatmap_data).T

    # keywords = list(freq_df.index) # Get keywords from the dataframe index
    # Your custom color palette
    neptune = '#86C3BC'
    astroblue = '#FD6626' #'#003D59'
    custom_colors = ['#003D59', '#FD6626', '#177070', '#FB871D', '#641B5E',
    ↪ '#86C3BC', '#F5E1FD', '#414A4F', 'k']

    # Create custom colormap
    custom_cmap = LinearSegmentedColormap.from_list('neptune_blue', [neptune,
    ↪ astroblue])

    # Create figure

```

```

plt.figure(figsize=(14, 8))

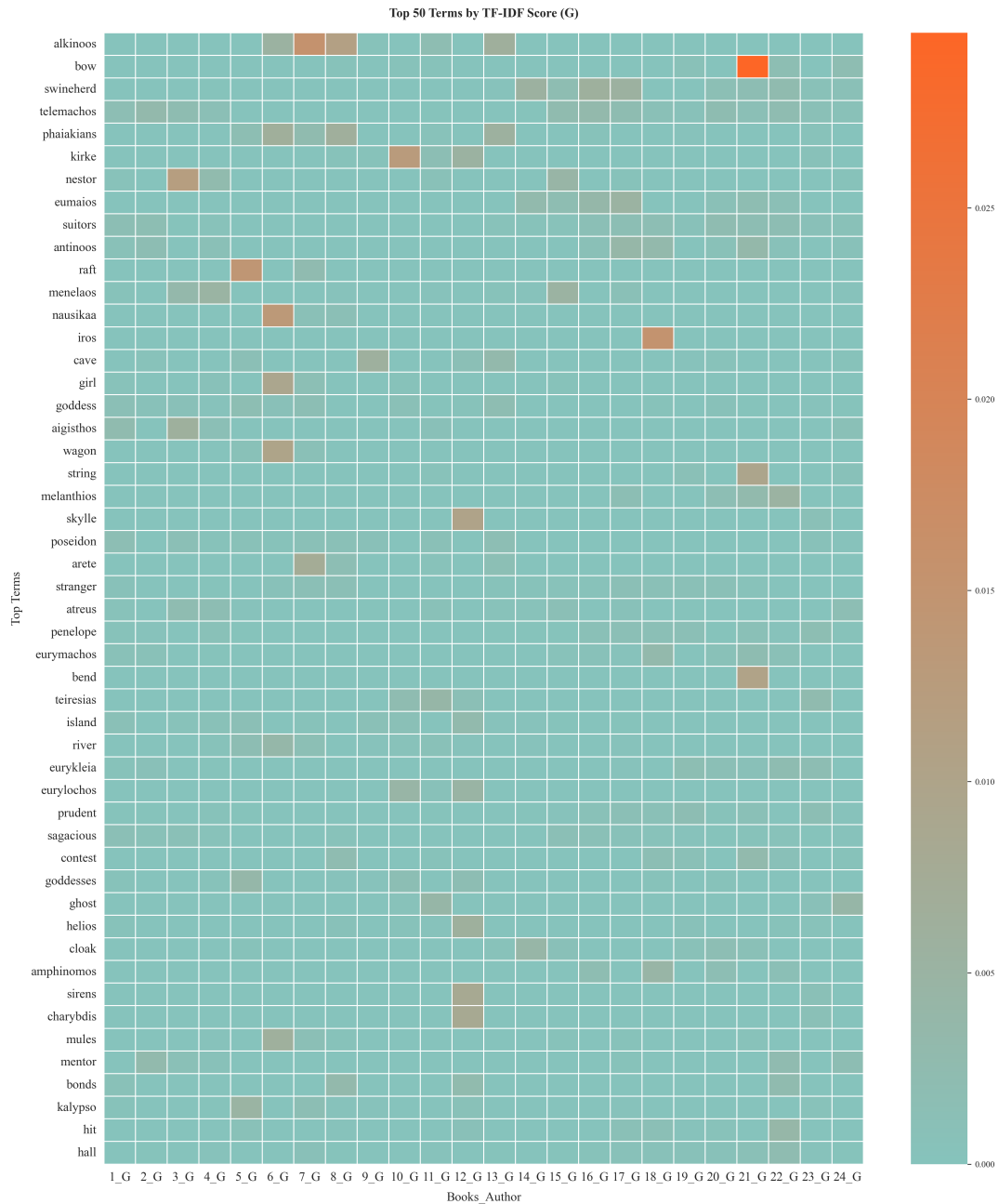
# Create heatmap
plt.figure(figsize=(14, 16))
sns.heatmap(heatmap_array, cmap=custom_cmap,
            xticklabels=book_ids, yticklabels=top_terms_list,
            cbar=True,
            linewidths=0.5,
            linecolor='white')
plt.title(f'Top {top_n} Terms by TF-IDF Score ({df["book_id"].iloc[0][2:
↪]])')
plt.xlabel('Books_Author')
plt.ylabel('Top Terms', fontsize=12, rotation=90)
plt.xticks(rotation=0, fontsize=12,)
plt.yticks(fontsize=12, rotation=0)
plt.tight_layout()
plt.savefig(f"/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
↪MVP-TFIDF_heatmap({df['book_id'].iloc[0][2:]}-{top_n}words).png")
plt.show()

return heatmap_array

top_terms_per_book_G, top_terms_overall_G = extract_top_terms(df_tfidf_G)
heatmap_array = create_tfidf_heatmap(df_tfidf_G)

```

<Figure size 4200x2400 with 0 Axes>

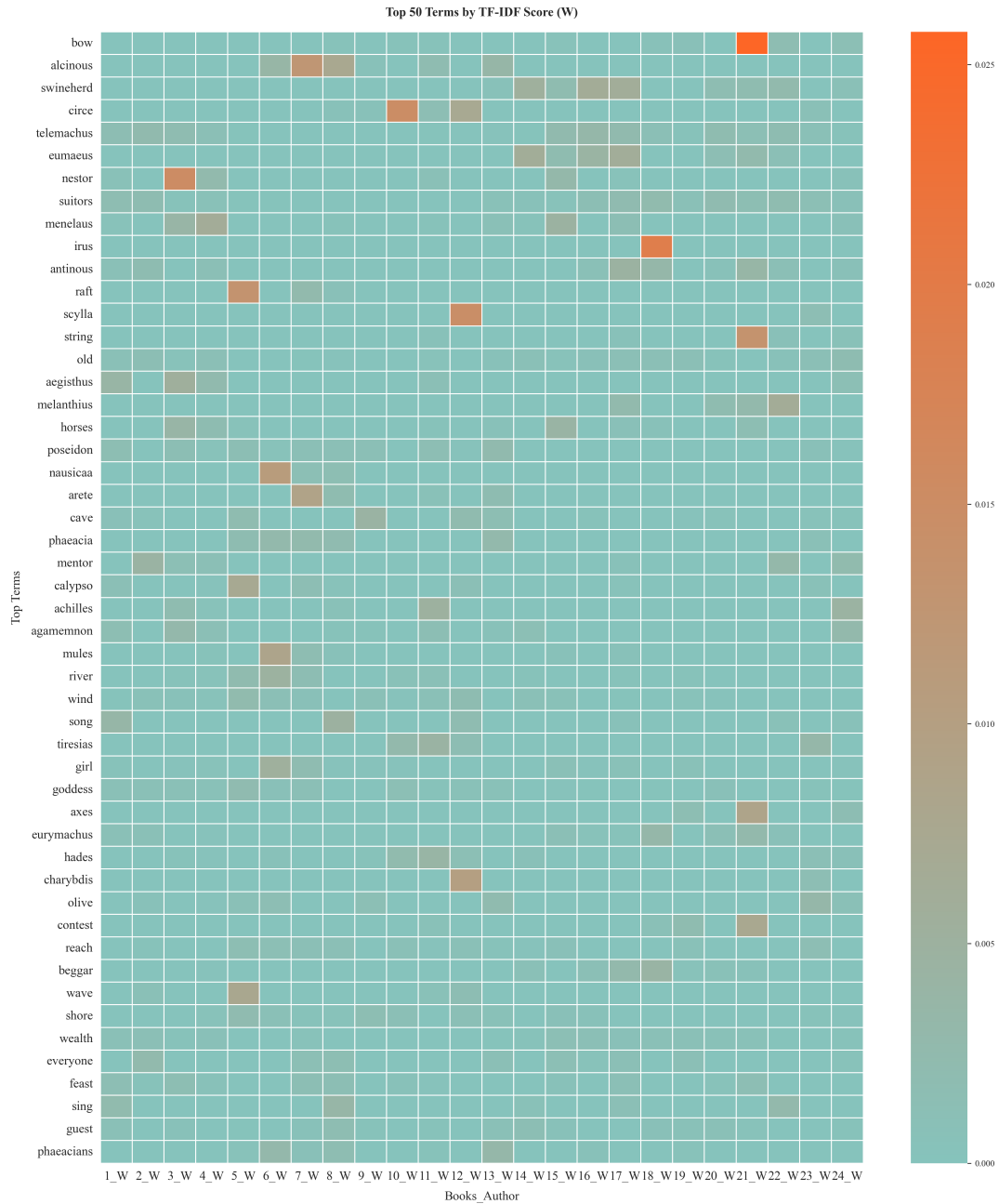


```
[24]: df_tfidf_W = calculate_tfidf(df_W)
df_tfidf_W[["book_id","tf_idf"]].head(2)
```

```
[24]: book_id      tf_idf
0      1_W  {'tell': -0.00033054246575105396, 'complicated...
1      2_W  {'dangerous': 0.000732546778414169, 'journey':...
```

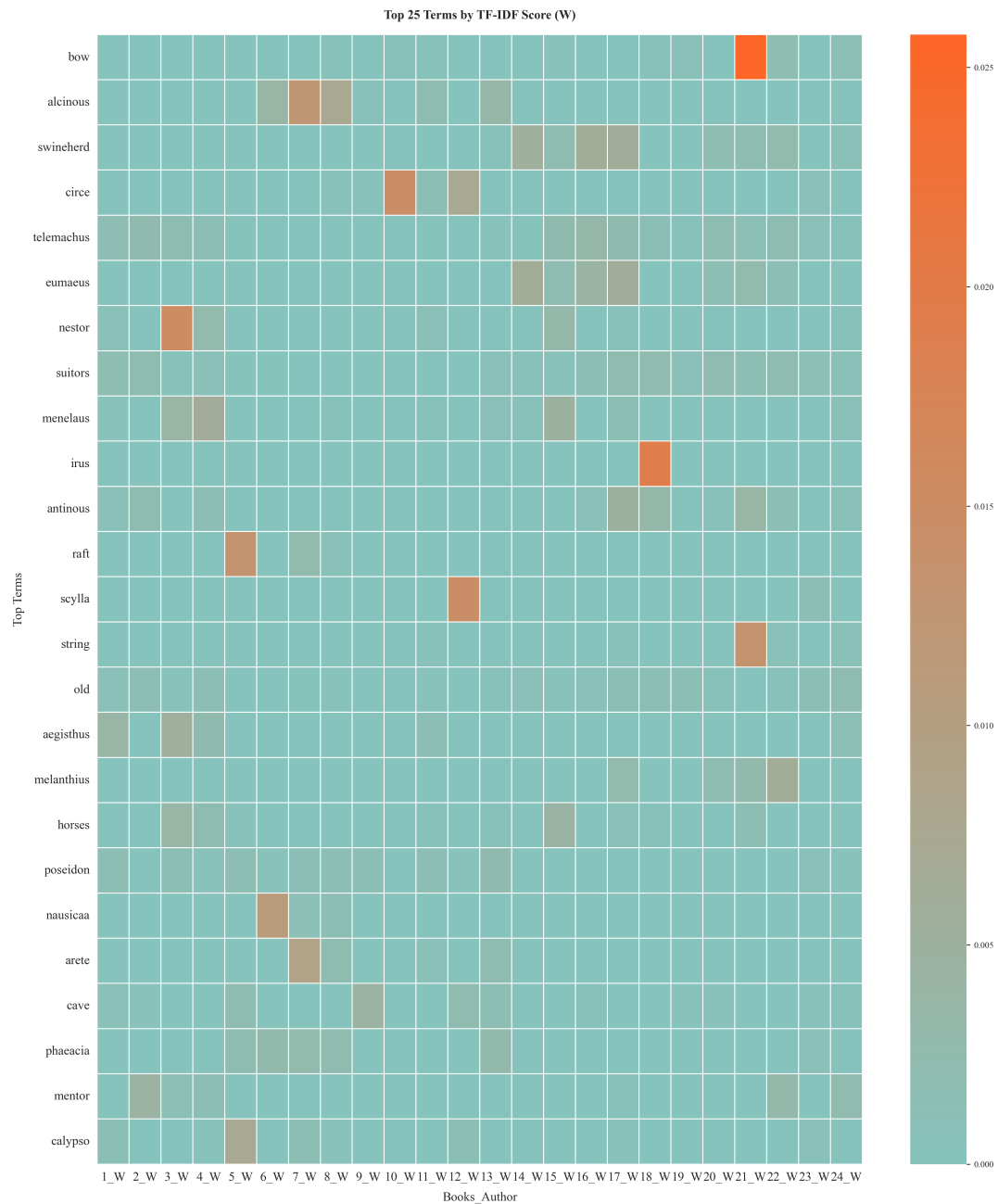
```
[25]: top_terms_per_book_W, top_terms_overall_W = extract_top_terms(df_W)
heatmap_array = create_tfidf_heatmap(df_tfidf_W)
```

<Figure size 4200x2400 with 0 Axes>



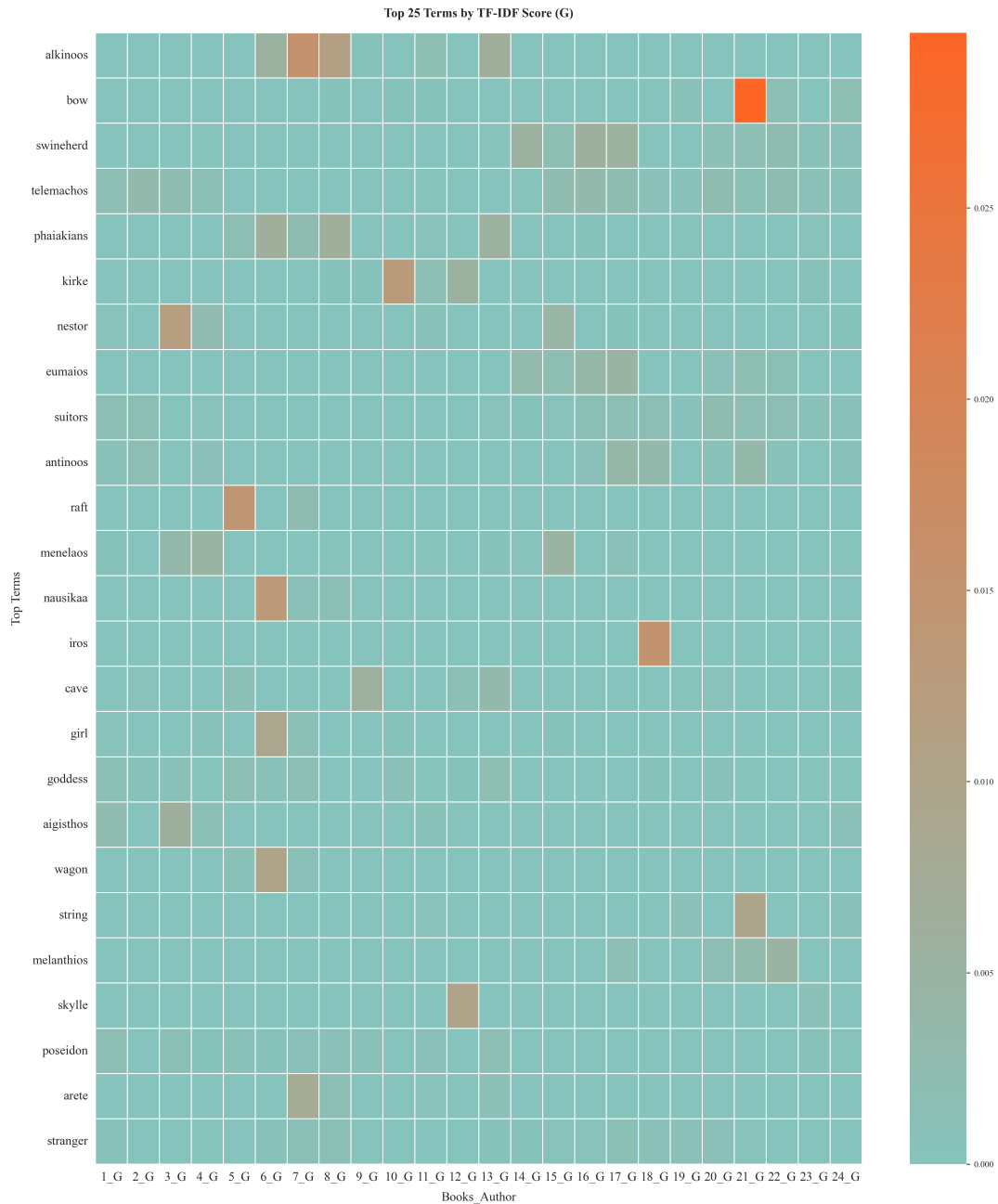
```
[26]: # 25 words for best clarity
heatmap_array = create_tfidf_heatmap(df_tfidf_W, top_n=25)
```


<Figure size 4200x2400 with 0 Axes>



```
[27]: # 25 words for best clarity
heatmap_array = create_tfidf_heatmap(df_tfidf_G, top_n=25)
```

<Figure size 4200x2400 with 0 Axes>



Wilson shows similar results to Green, using the step by step approach. Wilson's Odyssey (df) will be processed by the **e2e** function, so both translations have the same processing.

```
[28]: # Getting and then comparing W & G's
# 30 top terms
tt_W = top_terms_overall_W['term'][:30]
tt_G = top_terms_overall_G['term'][:30]
```

```
# New df with the top terms from both for comparison
top_terms_overall = pd.DataFrame({'Wilson': tt_W, 'Green': tt_G})
top_terms_overall.T
```

```
[28]:
```

	0	1	2	3	4	5	\
Wilson	bow	alcinous	swineherd	circe	telemachus	eumaeus	
Green	alkinoos	bow	swineherd	telemachos	phaiakians	kirke	

	6	7	8	9	...	20	21	\
Wilson	nestor	suitors	menelaus	irus	...	arete	cave	
Green	nestor	eumaios	suitors	antinoos	...	melanthios	skylle	

	22	23	24	25	26	27	28	\
Wilson	phaeacia	mentor	calypso	achilles	agamemnon	mules	river	
Green	poseidon	arete	stranger	atreus	penelope	eurymachos	bend	

	29
Wilson	wind
Green	teiresias

[2 rows x 30 columns]

- **Parametric Hypothesis Testing:** Assessing significant variations.
Checking at the word by word comparison the similarities in choice of words is apparent. Thus the next step will be testing, but before choosing the test we need to check if the data distribution is normal.

```
[29]: # Calculate the sum of the tf-idf scores for each translation
itidf_W = df_tfidf_W["tf_idf"].apply(lambda x: list(x.values())).sum()
itidf_G = df_tfidf_G["tf_idf"].apply(lambda x: list(x.values())).sum()
```

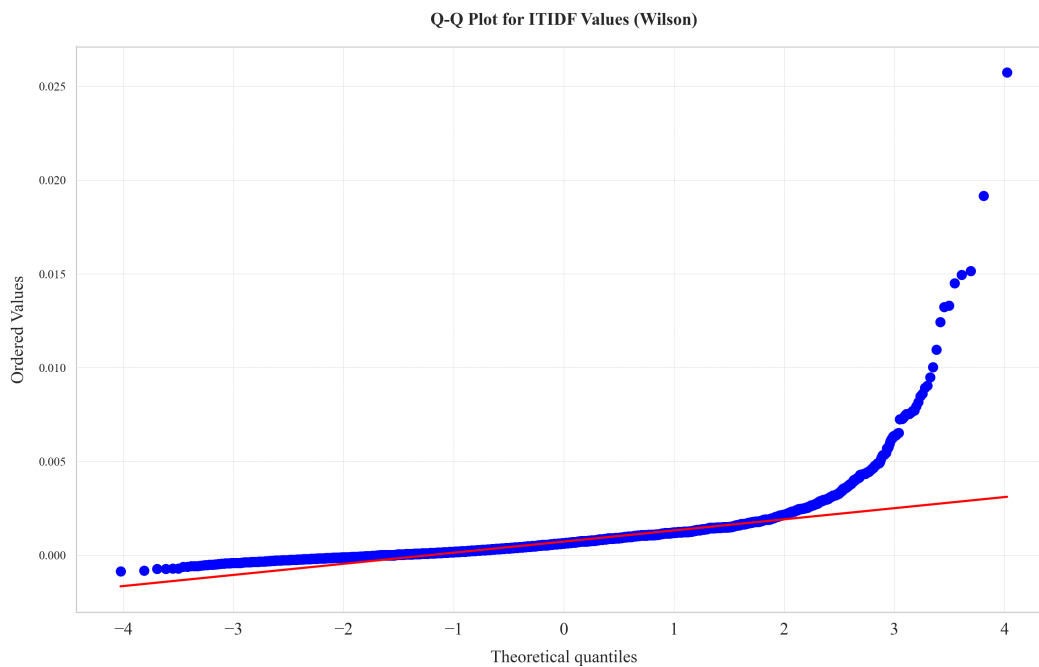
```
[30]: import scipy.stats as stats
# Q-Q plot for ITIDF values
def plot_qq(itidf, translator="Wilson"):
    """
    Create a Q-Q plot for the ITIDF values.

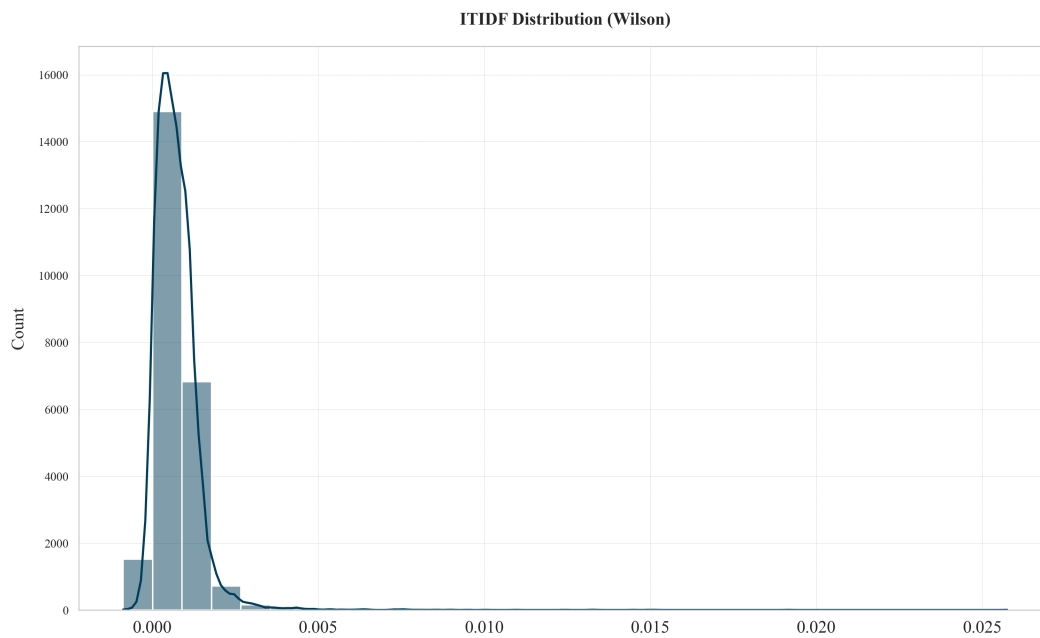
    Parameters:
    -----
    itidf : list
        List of ITIDF values to plot
    """
    # Generate a Q-Q plot
    stats.probplot(itidf_W, dist="norm", plot=plt)
    plt.title(f"Q-Q Plot for ITIDF Values ({translator})")
    plt.savefig(f"/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
    ↪Q-Q_Plot_{translator}.png")
    plt.show()
```

```
def plot_itidf_distribution(itidf, translator="Wilson"):
    """
    Plot the distribution of ITIDF values.

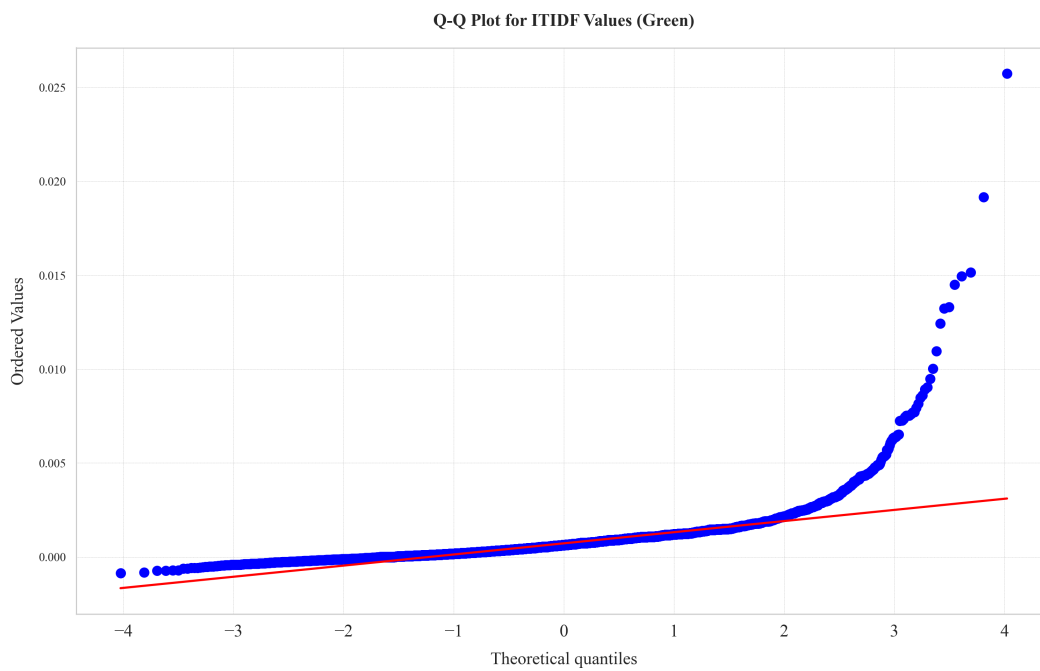
    Parameters:
    -----
    itidf : list
        List of ITIDF values to plot
    """
    # Create a histogram of the ITIDF values
    sns.histplot(itidf, bins=30, kde=True) # KDE adds a smoothed curve
    plt.title(f"ITIDF Distribution ({translator})")
    plt.savefig(f"/Users/debr/English-Homer/MVP_Green-Wilson/MVP_plots/
    ↪ITIDF_Distribution_{translator}.png")
    plt.show()

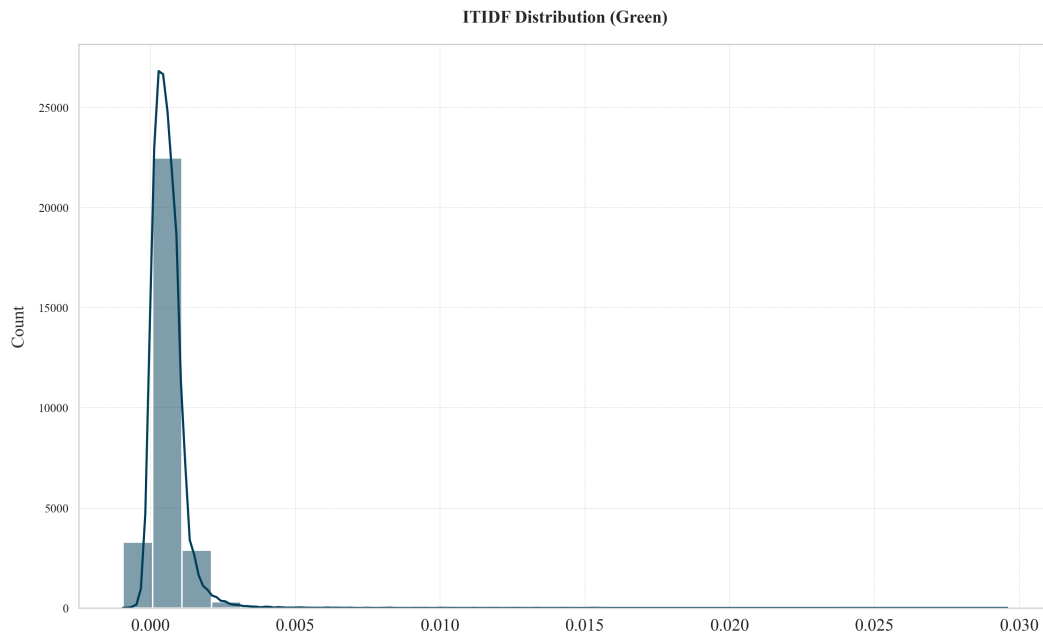
# Plot Q-Q plot and distribution for ITIDF values for Wilson
plot_qq(itidf_W, "Wilson")
plot_itidf_distribution(itidf_W, "Wilson")
```





```
[31]: # Plot Q-Q plot and distribution for ITIDF values for Green
plot_qq(itidf_G, "Green")
plot_itidf_distribution(itidf_G, "Green")
```





```
[32]: from scipy.stats import shapiro
# Shapiro test for normality
def shapiro_test(data):
    """
    Perform the Shapiro-Wilk test for normality from scipy.stats and print the
    results.

    Parameters:
    -----
    data : list or array
        The data to be tested for normality.

    Returns: shapiro statistic and p-value

    """
    stat, p = shapiro(data)
    print(f"Shapiro-Wilk Test: p-value = {p}")

    if p > 0.05:
        print("Data appears to be normally distributed (fail to reject H0).")
    else:
        print("Data is not normally distributed (reject H0).")
```

```

    return stat, p

# Shapiro-Wilk test for normality
print("Wilson")
shapiro_test(itidf_W)
print("\n")
print("Green")
shapiro_test(itidf_G)

```

Wilson

Shapiro-Wilk Test: p-value = 1.9264430476087096e-105
Data is not normally distributed (reject H0).

Green

Shapiro-Wilk Test: p-value = 1.5939662570478783e-113
Data is not normally distributed (reject H0).

```
[32]: (np.float64(0.6885523258795925), np.float64(1.5939662570478783e-113))
```

Why the Mann-Whitney U Test

Since the data for the TF-IDF values is not normally distributed, a parametric test like the t-test is not appropriate. Instead, the Mann-Whitney U test is used, which is a non-parametric test for comparing two independent samples. It does not require the assumption of normality and is suitable for comparing distributions when the data is skewed or has outliers.

Hypotheses:

- Null Hypothesis (H): There is no significant difference in the TF-IDF values between the two translations (Wilson and Green).
- Alternative Hypothesis (H): There is a significant difference in the TF-IDF values between the two translations.

The Mann-Whitney U test evaluates whether the distributions of TF-IDF values for the two translations differ, considering their ranks rather than specific values.

```

[33]: # Mannwhitney U test for comparing ITIDF values
from scipy.stats import mannwhitneyu # Import the actual function

def mannwhitneyu_test(x, y, alternative='two-sided'):
    """
    Perform the Mann-Whitney U test for comparing two independent samples.
    """
    stat, p = mannwhitneyu(x, y, alternative=alternative) # Use SciPy's
    ↪function
    print(f"Mann-Whitney U test statistic: {stat}, p-value: {p}")

    if p < 0.05:

```

```

        print("Reject H : The distributions of the translations are_
↪significantly different.")
    else:
        print("Fail to reject H : No significant difference between the_
↪translations.")

# Run the test
mannwhitneyu_test(itidf_W, itidf_G, alternative='two-sided')

```

Mann-Whitney U test statistic: 398451440.5, p-value: 2.3169071909419926e-146
Reject H : The distributions of the translations are significantly different.

3 Discussion of Results

After conducting three experiments with accompanying statistical tests, our analysis provides valuable insights into the lexical characteristics of Green and Wilson’s translations of *The Odyssey*. While our sample size represents an MVP analysis and cannot support definitive conclusions, the methodology demonstrates effectiveness in yielding meaningful linguistic insights.

3.1 Type-Token Ratio (TTR) Analysis

3.1.1 Statistical Findings

The t-test comparing TTR values between Wilson’s and Green’s translations revealed no statistically significant difference ($p > 0.05$), supporting the null hypothesis that their lexical diversity is quantitatively similar. Rather than being inconclusive, this finding offers critical insight into translation dynamics.

3.1.2 Linguistic Interpretation

The comparable TTR values suggest that both translators, despite working independently, operate within similar lexical parameters. This statistical similarity indicates that both translations maintain equivalent levels of vocabulary richness and complexity. The absence of significant difference points to:

- **Shared Cultural Framework:** Both translators draw from the same linguistic resources and cultural contexts, influencing their word choice patterns.
- **Similar Translation Approaches:** Despite potential stylistic differences, both translators appear to balance faithfulness to the source text with considerations of modern readability.
- **Common Professional Standards:** The comparable lexical diversity suggests adherence to similar academic and literary translation norms.

3.2 Zipf’s Law Analysis

3.2.1 Statistical Findings

The nearly identical 95% confidence intervals for Wilson’s and Green’s Zipfian slopes demonstrate that both translations follow remarkably similar word frequency distributions. This statistical congruence further validates the patterns observed in the TTR analysis.

3.2.2 Linguistic Interpretation

The Zipfian distribution results indicate that both translations maintain natural statistical language structures. Despite individual stylistic choices, both translators:

- **Preserve Natural Language Patterns:** Both translations adhere to the expected power law distribution of word frequencies that characterizes natural language.
- **Balance Common and Rare Words:** The similar slopes suggest comparable approaches to balancing frequently used words with more specialized vocabulary.
- **Reflect Similar Lexical Economy:** Both translations demonstrate equivalent efficiency in word usage patterns, suggesting similar approaches to conveying meaning within the constraints of English.

3.3 TF-IDF Analysis

3.3.1 Statistical Findings

The Mann-Whitney U test (test statistic: 398451440.5, p-value: 2.32e-146) revealed a highly significant difference between the TF-IDF distributions of the two translations. This extremely small p-value strongly rejects the null hypothesis of similar distributions.

3.3.2 Linguistic Interpretation

Unlike the previous two measures, the TF-IDF analysis reveals significant differences in how the two translators emphasize specific words and concepts:

- **Distinct Emphasis Patterns:** While overall lexical diversity is similar, the translators differ markedly in which specific words they emphasize relative to the overall corpus.
- **Unique Interpretive Approaches:** The divergent TF-IDF values suggest different interpretations of which concepts require emphasis or elaboration.
- **Different Lexical Focusing:** Despite similar overall vocabulary richness, the two translators appear to focus on different elements of the narrative through their word choice.

3.4 Implications for Translation Studies

These findings have several important implications for translation studies:

1. **Cultural Embeddedness of Translation:** The similarity in TTR and Zipfian distributions supports Bourdieu's concept of habitus—translators internalize shared linguistic and literary norms that influence their work, even unconsciously.
2. **Balancing Individual Style and Cultural Context:** While translators maintain individual approaches (evident in TF-IDF differences), they remain bound by shared linguistic and cultural forces (shown in TTR and Zipf similarities).
3. **Quantifiable Translation Differences:** The contrasting results between frequency-based measures (TTR, Zipf) and emphasis-based measures (TF-IDF) suggest that translation differences may be more evident in which elements translators choose to emphasize rather than in overall vocabulary richness—translator's agency.
4. **Methodological Insights:** The combination of these three statistical approaches provides a more nuanced understanding than any single measure could offer, demonstrating the value of multi-dimensional analysis in translation studies.

3.5 Future Research Directions

To build upon this MVP analysis, future research could explore:

3.5.1 More Methods:

- **Lexical Choice (Etymology) Analysis**
- **Syntactic Complexity Methods**
- **Stylometrics and Thematic Mapping**

3.5.2 Expanded Corpus:

Applying these methods to a larger sample of the translations to confirm whether these patterns persist throughout the complete works. And thus...

- **Cross-Temporal Studies:** Expanding the analysis to include translations from different time periods to determine how temporal context influences translation.

This quantitative analysis serves as an empirical foundation for qualitative exploration of how Green and Wilson have shaped *The Odyssey's* language, structure, and poetic essence, enriching our understanding of literary interpretation in classical texts.